$f(x) = x^3 - \cos x \qquad x_1 = 0 \quad \& \quad x_2 = 1$

Solve & compare Bisection & false position method

Bisection method

$x_3 = \frac{x_1 + x_2}{2}$

iteration	x_1	$f(x_1)$	x_2	$f(x_2)$	x_3	$f(x_3)$
1						
2						

```
function f(x: number)
{
    return (x*x*x - Math.cos(x))
}

for (i=1 ; i<=10 ; i++)
{
    x3 = (x1 + x2)/2
    xx1.push(x1)
    xx2.push(x2)
    xx3.push(x3)
    fx1.push(f(x1))
    fx2.push(f(x2))
    fx3.push(f(x3))

    If f(x1) x f(x3) < 1
    {
        x2 = x3;
    }
    else
    {   x1 = x3;
    }
}
```

$x_1 \rightarrow xx_1 = [\,]$

$fx_1 \rightarrow = [\,]$

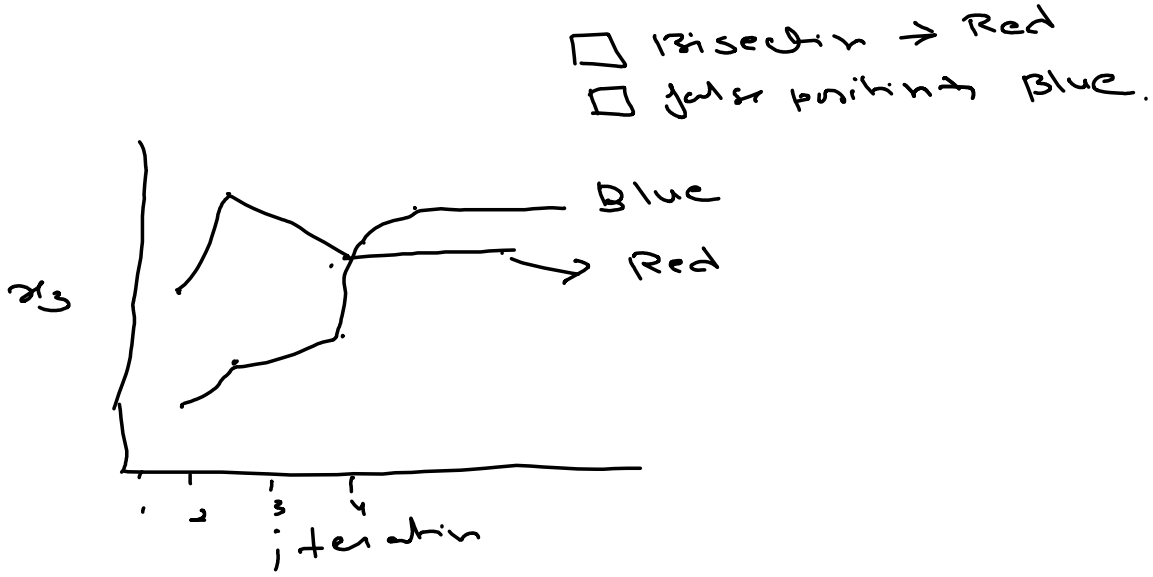
$xx_2 \rightarrow = [\,]$

$fx_2 \rightarrow = [\,]$

$xx_3 \rightarrow = [\,]$

$fx_3 \rightarrow = [\,]$

false position method

$x_3 =$

```
x3 = x1 - f(x1) * ((x2 - x1) / (f(x2) - f(x1)));
```